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Automated Report

Technical Report for

Aries Clean Energy

PFAS Analysis

SGS Job Number: FA69813

Sampling Dates: 10/16/19 - 10/30/19

Report to:

Aries Clean Energy
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ATTN: Matt Newman

Total number of pages in report: **26**



Test results contained within this data package meet the requirements of the National Environmental Laboratory Accreditation Program and/or state specific certification programs as applicable.

A handwritten signature in black ink that reads 'Caitlin Brice'.

Caitlin Brice, M.S.
General Manager

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Certifications: FL(E83510), LA(03051), KS(E-10327), IL(200063), NC(573), NJ(FL002), NY(12022), SC(96038001)
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Test results relate only to samples analyzed.

Table of Contents

-1-

Section 1: Sample Summary 3

Section 2: Summary of Hits 4

Section 3: Sample Results 5

3.1: FA69813-1: ASR FEED STOCK 6

3.2: FA69813-2: CYCLON/BED OXIDING 8

3.3: FA69813-3: REDUCING BED 10

3.4: FA69813-4: FILTER MATERIAL 12

Section 4: Misc. Forms 14

4.1: Certification Exceptions 15

4.2: Chain of Custody 16

Section 5: MS Semi-volatiles - QC Data Summaries 18

5.1: Method Blank Summary 19

5.2: Blank Spike Summary 23

5.3: Matrix Spike/Matrix Spike Duplicate Summary 25



Sample Summary

Aries Clean Energy

Job No: FA69813

PFAS Analysis

Sample Number	Collected Date	Time By	Received	Matrix Code	Type	Client Sample ID
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This report contains results reported as ND = Not detected. The following applies:
Organics ND = Not detected above the MDL

FA69813-1	10/30/19	00:00	MSMN11/12/19	SO	Solid	ASR FEED STOCK
FA69813-2	10/16/19	17:25	MSMN11/12/19	SO	Solid	CYCLON/BED OXIDING
FA69813-3	10/16/19	17:20	MSMN11/12/19	SO	Solid	REDUCING BED
FA69813-4	10/16/19	17:25	MSMN11/12/19	SO	Solid	FILTER MATERIAL

Soil samples reported on a dry weight basis unless otherwise indicated on result page.

Summary of Hits

Page 1 of 1

Job Number: FA69813
Account: Aries Clean Energy
Project: PFAS Analysis
Collected: 10/16/19 thru 10/30/19

Lab Sample ID	Client Sample ID	Result/ Qual	RL	MDL	Units	Method
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FA69813-1 ASR FEED STOCK

Perfluorobutanoic acid	0.554 J	1.0	0.26	ug/kg	EPA 537M BY ID
Perfluorohexanoic acid	0.633 J	1.0	0.21	ug/kg	EPA 537M BY ID
Perfluorobutanesulfonic acid	1.49	1.0	0.26	ug/kg	EPA 537M BY ID
Perfluorohexanesulfonic acid	0.390 J	1.0	0.26	ug/kg	EPA 537M BY ID
Perfluorooctanesulfonic acid	4.32	1.0	0.26	ug/kg	EPA 537M BY ID
EtFOSAA	0.788 J	2.6	0.52	ug/kg	EPA 537M BY ID
6:2 Fluorotelomer sulfonate	0.946 J	1.0	0.26	ug/kg	EPA 537M BY ID

FA69813-2 CYCLON/BED OXIDING

No hits reported in this sample.

FA69813-3 REDUCING BED

No hits reported in this sample.

FA69813-4 FILTER MATERIAL

Perfluorobutanoic acid	0.387 J	1.0	0.25	ug/kg	EPA 537M BY ID
Perfluoropentanoic acid	1.12	1.0	0.20	ug/kg	EPA 537M BY ID

Sample Results

Report of Analysis

Report of Analysis

Client Sample ID:	ASR FEED STOCK	Date Sampled:	10/30/19
Lab Sample ID:	FA69813-1	Date Received:	11/12/19
Matrix:	SO - Solid	Percent Solids:	96.3
Method:	EPA 537M BY ID IN HOUSE		
Project:	PFAS Analysis		

	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
Run #1	2Q39355.D	1	11/14/19 09:02	NAF	11/12/19 16:30	OP77685	S2Q605
Run #2 ^a	2Q39323.D	10	11/14/19 00:21	NAF	11/12/19 16:30	OP77685	S2Q605

	Initial Weight	Final Volume
Run #1	1.99 g	1.0 ml
Run #2	1.99 g	1.0 ml

CAS No.	Compound	Result	RL	MDL	Units	Q
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PERFLUOROALKYLCARBOXYLIC ACIDS

375-22-4	Perfluorobutanoic acid	0.554	1.0	0.26	ug/kg	J
2706-90-3	Perfluoropentanoic acid	ND	1.0	0.21	ug/kg	
307-24-4	Perfluorohexanoic acid	0.633	1.0	0.21	ug/kg	J
375-85-9	Perfluoroheptanoic acid	ND ^b	10	2.6	ug/kg	
335-67-1	Perfluorooctanoic acid	ND ^b	10	2.6	ug/kg	
375-95-1	Perfluorononanoic acid	ND ^b	10	2.6	ug/kg	
335-76-2	Perfluorodecanoic acid	ND ^b	10	2.6	ug/kg	
2058-94-8	Perfluoroundecanoic acid	ND ^b	10	2.6	ug/kg	
307-55-1	Perfluorododecanoic acid	ND ^b	10	2.6	ug/kg	
72629-94-8	Perfluorotridecanoic acid	ND	1.0	0.26	ug/kg	
376-06-7	Perfluorotetradecanoic acid	ND	1.0	0.26	ug/kg	

PERFLUOROALKYLSULFONATES

375-73-5	Perfluorobutanesulfonic acid	1.49	1.0	0.26	ug/kg	
2706-91-4	Perfluoropentanesulfonic acid	ND	1.0	0.26	ug/kg	
355-46-4	Perfluorohexanesulfonic acid	0.390	1.0	0.26	ug/kg	J
375-92-8	Perfluoroheptanesulfonic acid	ND	1.0	0.26	ug/kg	
1763-23-1	Perfluorooctanesulfonic acid	4.32	1.0	0.26	ug/kg	
68259-12-1	Perfluorononanesulfonic acid	ND	1.0	0.26	ug/kg	
335-77-3	Perfluorodecanesulfonic acid	ND	1.0	0.26	ug/kg	

PERFLUOROCTANESULFONAMIDES

754-91-6	PFOSA ^c	ND ^b	10	2.6	ug/kg	
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PERFLUOROCTANESULFONAMIDOACETIC ACIDS

2355-31-9	MeFOSAA	ND	2.6	0.52	ug/kg	
2991-50-6	EtFOSAA	0.788	2.6	0.52	ug/kg	J

FLUOROTELOMER SULFONATES

757124-72-4	4:2 Fluorotelomer sulfonate ^c	ND ^b	10	2.6	ug/kg	
27619-97-2	6:2 Fluorotelomer sulfonate	0.946	1.0	0.26	ug/kg	J

ND = Not detected MDL = Method Detection Limit

RL = Reporting Limit

E = Indicates value exceeds calibration range

J = Indicates an estimated value

B = Indicates analyte found in associated method blank

N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	ASR FEED STOCK	Date Sampled:	10/30/19
Lab Sample ID:	FA69813-1	Date Received:	11/12/19
Matrix:	SO - Solid	Percent Solids:	96.3
Method:	EPA 537M BY ID IN HOUSE		
Project:	PFAS Analysis		

CAS No.	Compound	Result	RL	MDL	Units	Q
39108-34-4	8:2 Fluorotelomer sulfonate	ND	1.0	0.26	ug/kg	

NEXT GENERATION PFAS ANALYTES

13252-13-6	HFPO-DA (GenX)	ND	5.2	1.3	ug/kg	
919005-14-4	ADONA	ND	1.0	0.26	ug/kg	
756426-58-1	9Cl-PF3ONS (F-53B Major)	ND	1.0	0.26	ug/kg	
763051-92-9	11Cl-PF3OUdS (F-53B Minor)	ND	1.0	0.26	ug/kg	

CAS No.	ID Standard Recoveries	Run# 1	Run# 2	Limits
	13C4-PFBA	55%	54%	50-150%
	13C5-PFPeA	55%	54%	50-150%
	13C5-PFHxA	51%	50%	50-150%
	13C4-PFHpA	47% ^d	51%	50-150%
	13C8-PFOA	48% ^d	51%	50-150%
	13C9-PFNA	49% ^d	53%	50-150%
	13C6-PFDA	45% ^d	52%	50-150%
	13C7-PFUnDA	29% ^d	48% ^d	50-150%
	13C2-PFDoDA	29% ^d	53%	50-150%
	13C2-PFTeDA	63%	57%	50-150%
	13C3-PFBS	55%	55%	50-150%
	13C3-PFHxS	51%	54%	50-150%
	13C8-PFOS	51%	56%	50-150%
	13C8-FOSA	30% ^d	51%	50-150%
	d3-MeFOSAA	56%	57%	50-150%
	13C2-4:2FTS	49% ^d	48% ^d	50-150%
	13C2-6:2FTS	53%	56%	50-150%
	13C2-8:2FTS	54%	64%	50-150%
	13C3-HFPO-DA	62%	56%	50-150%

(a) Dilution required due to matrix interference (internal standard failure).

(b) Result is from Run# 2

(c) Associated ID Standard outside control limits due to matrix interference. Confirmed by reanalysis.

(d) Outside control limits due to matrix interference.

ND = Not detected MDL = Method Detection Limit

RL = Reporting Limit

E = Indicates value exceeds calibration range

J = Indicates an estimated value

B = Indicates analyte found in associated method blank

N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	CYCLON/BED OXIDING	
Lab Sample ID:	FA69813-2	Date Sampled: 10/16/19
Matrix:	SO - Solid	Date Received: 11/12/19
Method:	EPA 537M BY ID IN HOUSE	Percent Solids: 99.9
Project:	PFAS Analysis	

	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
Run #1	2Q39356.D	1	11/14/19 09:19	NAF	11/12/19 16:30	OP77685	S2Q605
Run #2							

	Initial Weight	Final Volume
Run #1	2.02 g	1.0 ml
Run #2		

CAS No.	Compound	Result	RL	MDL	Units	Q
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PERFLUOROALKYLCARBOXYLIC ACIDS

375-22-4	Perfluorobutanoic acid	ND	0.99	0.25	ug/kg	
2706-90-3	Perfluoropentanoic acid	ND	0.99	0.20	ug/kg	
307-24-4	Perfluorohexanoic acid	ND	0.99	0.20	ug/kg	
375-85-9	Perfluoroheptanoic acid	ND	0.99	0.25	ug/kg	
335-67-1	Perfluorooctanoic acid	ND	0.99	0.25	ug/kg	
375-95-1	Perfluorononanoic acid	ND	0.99	0.25	ug/kg	
335-76-2	Perfluorodecanoic acid	ND	0.99	0.25	ug/kg	
2058-94-8	Perfluoroundecanoic acid	ND	0.99	0.25	ug/kg	
307-55-1	Perfluorododecanoic acid	ND	0.99	0.25	ug/kg	
72629-94-8	Perfluorotridecanoic acid	ND	0.99	0.25	ug/kg	
376-06-7	Perfluorotetradecanoic acid	ND	0.99	0.25	ug/kg	

PERFLUOROALKYLSULFONATES

375-73-5	Perfluorobutanesulfonic acid	ND	0.99	0.25	ug/kg	
2706-91-4	Perfluoropentanesulfonic acid	ND	0.99	0.25	ug/kg	
355-46-4	Perfluorohexanesulfonic acid	ND	0.99	0.25	ug/kg	
375-92-8	Perfluoroheptanesulfonic acid	ND	0.99	0.25	ug/kg	
1763-23-1	Perfluorooctanesulfonic acid	ND	0.99	0.25	ug/kg	
68259-12-1	Perfluorononanesulfonic acid	ND	0.99	0.25	ug/kg	
335-77-3	Perfluorodecanesulfonic acid	ND	0.99	0.25	ug/kg	

PERFLUOROCTANESULFONAMIDES

754-91-6	PFOSA	ND	0.99	0.25	ug/kg	
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PERFLUOROCTANESULFONAMIDOACETIC ACIDS

2355-31-9	MeFOSAA	ND	2.5	0.50	ug/kg	
2991-50-6	EtFOSAA	ND	2.5	0.50	ug/kg	

FLUOROTELOMER SULFONATES

757124-72-4	4:2 Fluorotelomer sulfonate	ND	0.99	0.25	ug/kg	
27619-97-2	6:2 Fluorotelomer sulfonate	ND	0.99	0.25	ug/kg	

ND = Not detected MDL = Method Detection Limit

RL = Reporting Limit

E = Indicates value exceeds calibration range

J = Indicates an estimated value

B = Indicates analyte found in associated method blank

N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	CYCLON/BED OXIDING	Date Sampled:	10/16/19
Lab Sample ID:	FA69813-2	Date Received:	11/12/19
Matrix:	SO - Solid	Percent Solids:	99.9
Method:	EPA 537M BY ID IN HOUSE		
Project:	PFAS Analysis		

CAS No.	Compound	Result	RL	MDL	Units	Q
39108-34-4	8:2 Fluorotelomer sulfonate	ND	0.99	0.25	ug/kg	

NEXT GENERATION PFAS ANALYTES

13252-13-6	HFPO-DA (GenX)	ND	5.0	1.2	ug/kg	
919005-14-4	ADONA	ND	0.99	0.25	ug/kg	
756426-58-1	9Cl-PF3ONS (F-53B Major)	ND	0.99	0.25	ug/kg	
763051-92-9	11Cl-PF3OUdS (F-53B Minor)	ND	0.99	0.25	ug/kg	

CAS No.	ID Standard Recoveries	Run# 1	Run# 2	Limits
	13C4-PFBA	65%		50-150%
	13C5-PFPeA	72%		50-150%
	13C5-PFHxA	69%		50-150%
	13C4-PFHpA	69%		50-150%
	13C8-PFOA	73%		50-150%
	13C9-PFNA	72%		50-150%
	13C6-PFDA	72%		50-150%
	13C7-PFUnDA	74%		50-150%
	13C2-PFDoDA	72%		50-150%
	13C2-PFTeDA	74%		50-150%
	13C3-PFBS	73%		50-150%
	13C3-PFHxS	72%		50-150%
	13C8-PFOS	75%		50-150%
	13C8-FOSA	74%		50-150%
	d3-MeFOSAA	68%		50-150%
	13C2-4:2FTS	63%		50-150%
	13C2-6:2FTS	70%		50-150%
	13C2-8:2FTS	73%		50-150%
	13C3-HFPO-DA	86%		50-150%

ND = Not detected MDL = Method Detection Limit
 RL = Reporting Limit
 E = Indicates value exceeds calibration range

J = Indicates an estimated value
 B = Indicates analyte found in associated method blank
 N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	REDUCING BED	Date Sampled:	10/16/19
Lab Sample ID:	FA69813-3	Date Received:	11/12/19
Matrix:	SO - Solid	Percent Solids:	100.0
Method:	EPA 537M BY ID IN HOUSE		
Project:	PFAS Analysis		

	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
Run #1	2Q39357.D	1	11/14/19 09:35	NAF	11/12/19 16:30	OP77685	S2Q605
Run #2							

	Initial Weight	Final Volume
Run #1	2.05 g	1.0 ml
Run #2		

CAS No.	Compound	Result	RL	MDL	Units	Q
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PERFLUOROALKYLCARBOXYLIC ACIDS

375-22-4	Perfluorobutanoic acid	ND	0.98	0.24	ug/kg	
2706-90-3	Perfluoropentanoic acid	ND	0.98	0.20	ug/kg	
307-24-4	Perfluorohexanoic acid	ND	0.98	0.20	ug/kg	
375-85-9	Perfluoroheptanoic acid	ND	0.98	0.24	ug/kg	
335-67-1	Perfluorooctanoic acid	ND	0.98	0.24	ug/kg	
375-95-1	Perfluorononanoic acid	ND	0.98	0.24	ug/kg	
335-76-2	Perfluorodecanoic acid	ND	0.98	0.24	ug/kg	
2058-94-8	Perfluoroundecanoic acid	ND	0.98	0.24	ug/kg	
307-55-1	Perfluorododecanoic acid	ND	0.98	0.24	ug/kg	
72629-94-8	Perfluorotridecanoic acid	ND	0.98	0.24	ug/kg	
376-06-7	Perfluorotetradecanoic acid	ND	0.98	0.24	ug/kg	

PERFLUOROALKYLSULFONATES

375-73-5	Perfluorobutanesulfonic acid	ND	0.98	0.24	ug/kg	
2706-91-4	Perfluoropentanesulfonic acid	ND	0.98	0.24	ug/kg	
355-46-4	Perfluorohexanesulfonic acid	ND	0.98	0.24	ug/kg	
375-92-8	Perfluoroheptanesulfonic acid	ND	0.98	0.24	ug/kg	
1763-23-1	Perfluorooctanesulfonic acid	ND	0.98	0.24	ug/kg	
68259-12-1	Perfluorononanesulfonic acid	ND	0.98	0.24	ug/kg	
335-77-3	Perfluorodecanesulfonic acid	ND	0.98	0.24	ug/kg	

PERFLUOROCTANESULFONAMIDES

754-91-6	PFOSA	ND	0.98	0.24	ug/kg	
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PERFLUOROCTANESULFONAMIDOACETIC ACIDS

2355-31-9	MeFOSAA	ND	2.4	0.49	ug/kg	
2991-50-6	EtFOSAA	ND	2.4	0.49	ug/kg	

FLUOROTELOMER SULFONATES

757124-72-4	4:2 Fluorotelomer sulfonate	ND	0.98	0.24	ug/kg	
27619-97-2	6:2 Fluorotelomer sulfonate	ND	0.98	0.24	ug/kg	

ND = Not detected MDL = Method Detection Limit

RL = Reporting Limit

E = Indicates value exceeds calibration range

J = Indicates an estimated value

B = Indicates analyte found in associated method blank

N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	REDUCING BED		
Lab Sample ID:	FA69813-3	Date Sampled:	10/16/19
Matrix:	SO - Solid	Date Received:	11/12/19
Method:	EPA 537M BY ID IN HOUSE	Percent Solids:	100.0
Project:	PFAS Analysis		

CAS No.	Compound	Result	RL	MDL	Units	Q
39108-34-4	8:2 Fluorotelomer sulfonate	ND	0.98	0.24	ug/kg	

NEXT GENERATION PFAS ANALYTES

13252-13-6	HFPO-DA (GenX)	ND	4.9	1.2	ug/kg	
919005-14-4	ADONA	ND	0.98	0.24	ug/kg	
756426-58-1	9Cl-PF3ONS (F-53B Major)	ND	0.98	0.24	ug/kg	
763051-92-9	11Cl-PF3OUdS (F-53B Minor)	ND	0.98	0.24	ug/kg	

CAS No.	ID Standard Recoveries	Run# 1	Run# 2	Limits
	13C4-PFBA	77%		50-150%
	13C5-PFPeA	79%		50-150%
	13C5-PFHxA	76%		50-150%
	13C4-PFHpA	76%		50-150%
	13C8-PFOA	80%		50-150%
	13C9-PFNA	80%		50-150%
	13C6-PFDA	78%		50-150%
	13C7-PFUnDA	72%		50-150%
	13C2-PFDoDA	74%		50-150%
	13C2-PFTeDA	69%		50-150%
	13C3-PFBS	81%		50-150%
	13C3-PFHxS	79%		50-150%
	13C8-PFOS	78%		50-150%
	13C8-FOSA	80%		50-150%
	d3-MeFOSAA	71%		50-150%
	13C2-4:2FTS	69%		50-150%
	13C2-6:2FTS	75%		50-150%
	13C2-8:2FTS	75%		50-150%
	13C3-HFPO-DA	97%		50-150%

ND = Not detected MDL = Method Detection Limit
 RL = Reporting Limit
 E = Indicates value exceeds calibration range

J = Indicates an estimated value
 B = Indicates analyte found in associated method blank
 N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	FILTER MATERIAL	
Lab Sample ID:	FA69813-4	Date Sampled: 10/16/19
Matrix:	SO - Solid	Date Received: 11/12/19
Method:	EPA 537M BY ID IN HOUSE	Percent Solids: 99.3
Project:	PFAS Analysis	

	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
Run #1	2Q39358.D	1	11/14/19 09:51	NAF	11/12/19 16:30	OP77685	S2Q605
Run #2 ^a	2Q39326.D	10	11/14/19 01:10	NAF	11/12/19 16:30	OP77685	S2Q605

	Initial Weight	Final Volume
Run #1	1.98 g	1.0 ml
Run #2	1.98 g	1.0 ml

CAS No.	Compound	Result	RL	MDL	Units	Q
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PERFLUOROALKYLCARBOXYLIC ACIDS

375-22-4	Perfluorobutanoic acid	0.387	1.0	0.25	ug/kg	J
2706-90-3	Perfluoropentanoic acid	1.12	1.0	0.20	ug/kg	
307-24-4	Perfluorohexanoic acid	ND	1.0	0.20	ug/kg	
375-85-9	Perfluoroheptanoic acid	ND	1.0	0.25	ug/kg	
335-67-1	Perfluorooctanoic acid	ND	1.0	0.25	ug/kg	
375-95-1	Perfluorononanoic acid	ND	1.0	0.25	ug/kg	
335-76-2	Perfluorodecanoic acid	ND	1.0	0.25	ug/kg	
2058-94-8	Perfluoroundecanoic acid	ND	1.0	0.25	ug/kg	
307-55-1	Perfluorododecanoic acid	ND	1.0	0.25	ug/kg	
72629-94-8	Perfluorotridecanoic acid	ND	1.0	0.25	ug/kg	
376-06-7	Perfluorotetradecanoic acid	ND	1.0	0.25	ug/kg	

PERFLUOROALKYLSULFONATES

375-73-5	Perfluorobutanesulfonic acid	ND	1.0	0.25	ug/kg	
2706-91-4	Perfluoropentanesulfonic acid	ND	1.0	0.25	ug/kg	
355-46-4	Perfluorohexanesulfonic acid	ND	1.0	0.25	ug/kg	
375-92-8	Perfluoroheptanesulfonic acid	ND	1.0	0.25	ug/kg	
1763-23-1	Perfluorooctanesulfonic acid	ND	1.0	0.25	ug/kg	
68259-12-1	Perfluorononanesulfonic acid	ND	1.0	0.25	ug/kg	
335-77-3	Perfluorodecanesulfonic acid	ND	1.0	0.25	ug/kg	

PERFLUOROOCCTANESULFONAMIDES

754-91-6	PFOSA ^b	ND ^c	10	2.5	ug/kg	
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PERFLUOROOCCTANESULFONAMIDOACETIC ACIDS

2355-31-9	MeFOSAA	ND	2.5	0.51	ug/kg	
2991-50-6	EtFOSAA	ND	2.5	0.51	ug/kg	

FLUOROTELOMER SULFONATES

757124-72-4	4:2 Fluorotelomer sulfonate ^b	ND ^c	10	2.5	ug/kg	
27619-97-2	6:2 Fluorotelomer sulfonate	ND	1.0	0.25	ug/kg	

ND = Not detected MDL = Method Detection Limit

RL = Reporting Limit

E = Indicates value exceeds calibration range

J = Indicates an estimated value

B = Indicates analyte found in associated method blank

N = Indicates presumptive evidence of a compound

Report of Analysis

Client Sample ID:	FILTER MATERIAL		
Lab Sample ID:	FA69813-4	Date Sampled:	10/16/19
Matrix:	SO - Solid	Date Received:	11/12/19
Method:	EPA 537M BY ID IN HOUSE	Percent Solids:	99.3
Project:	PFAS Analysis		

CAS No.	Compound	Result	RL	MDL	Units	Q
39108-34-4	8:2 Fluorotelomer sulfonate	ND	1.0	0.25	ug/kg	

NEXT GENERATION PFAS ANALYTES

13252-13-6	HFPO-DA (GenX)	ND	5.1	1.3	ug/kg	
919005-14-4	ADONA	ND	1.0	0.25	ug/kg	
756426-58-1	9Cl-PF3ONS (F-53B Major)	ND	1.0	0.25	ug/kg	
763051-92-9	11Cl-PF3OUdS (F-53B Minor)	ND	1.0	0.25	ug/kg	

CAS No.	ID Standard Recoveries	Run# 1	Run# 2	Limits
	13C4-PFBA	55%	53%	50-150%
	13C5-PFPeA	57%	54%	50-150%
	13C5-PFHxA	55%	51%	50-150%
	13C4-PFHpA	56%	50%	50-150%
	13C8-PFOA	61%	54%	50-150%
	13C9-PFNA	62%	54%	50-150%
	13C6-PFDA	60%	53%	50-150%
	13C7-PFUnDA	59%	55%	50-150%
	13C2-PFDoDA	61%	51%	50-150%
	13C2-PFTeDA	62%	51%	50-150%
	13C3-PFBS	60%	56%	50-150%
	13C3-PFHxS	59%	56%	50-150%
	13C8-PFOS	59%	57%	50-150%
	13C8-FOSA	46% ^d	49% ^d	50-150%
	d3-MeFOSAA	58%	52%	50-150%
	13C2-4:2FTS	47% ^d	45% ^d	50-150%
	13C2-6:2FTS	58%	51%	50-150%
	13C2-8:2FTS	74%	53%	50-150%
	13C3-HFPO-DA	74%	61%	50-150%

(a) Dilution required due to matrix interference (internal standard failure).

(b) Associated ID Standard outside control limits due to matrix interference. Confirmed by reanalysis.

(c) Result is from Run# 2

(d) Outside control limits due to matrix interference.

ND = Not detected MDL = Method Detection Limit

RL = Reporting Limit

E = Indicates value exceeds calibration range

J = Indicates an estimated value

B = Indicates analyte found in associated method blank

N = Indicates presumptive evidence of a compound

Misc. Forms

Custody Documents and Other Forms

Includes the following where applicable:

- Certification Exceptions
- Chain of Custody

Parameter Certification Exceptions

Page 1 of 1

Job Number: FA69813

Account: ARIESTNF Aries Clean Energy

Project: PFAS Analysis

The following parameters included in this report are exceptions to NELAC certification.
The certification status of each is indicated below.

Parameter	CAS#	Method	Mat	Certification Status
4:2 Fluorotelomer sulfonate	757124-72-4	EPA 537M BY ID	SO	Certified by SOP MS014
6:2 Fluorotelomer sulfonate	27619-97-2	EPA 537M BY ID	SO	Certified by SOP MS014
8:2 Fluorotelomer sulfonate	39108-34-4	EPA 537M BY ID	SO	Certified by SOP MS014
ADONA	919005-14-4	EPA 537M BY ID	SO	Certified by SOP MS014
11Cl-PF3OUdS (F-53B Minor)	763051-92-9	EPA 537M BY ID	SO	Certified by SOP MS014
9Cl-PF3ONS (F-53B Major)	756426-58-1	EPA 537M BY ID	SO	Certified by SOP MS014
EtFOSAA	2991-50-6	EPA 537M BY ID	SO	Certified by SOP MS014
HFPO-DA (GenX)	13252-13-6	EPA 537M BY ID	SO	Certified by SOP MS014
MeFOSAA	2355-31-9	EPA 537M BY ID	SO	Certified by SOP MS014
PFOSA	754-91-6	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorobutanesulfonic acid	375-73-5	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorobutanoic acid	375-22-4	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorodecanesulfonic acid	335-77-3	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorodecanoic acid	335-76-2	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorododecanoic acid	307-55-1	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluoroheptanesulfonic acid	375-92-8	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluoroheptanoic acid	375-85-9	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorohexanesulfonic acid	355-46-4	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorohexanoic acid	307-24-4	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorononanesulfonic acid	68259-12-1	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorononanoic acid	375-95-1	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorooctanesulfonic acid	1763-23-1	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorooctanoic acid	335-67-1	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluoropentanesulfonic acid	2706-91-4	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluoropentanoic acid	2706-90-3	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorotetradecanoic acid	376-06-7	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluorotridecanoic acid	72629-94-8	EPA 537M BY ID	SO	Certified by SOP MS014
Perfluoroundecanoic acid	2058-94-8	EPA 537M BY ID	SO	Certified by SOP MS014



SGS North America Inc - Orlando

Chain of Custody

4405 Vineland Road, Suite C-15 Orlando, FL 32811
TEL: 407-425-6700 FAX: 407-425-0707
www.sgs.com

SGS - ORLANDO JOB #:

PAGE OF

FA69813

Client / Reporting Information		Project Information		Analytical Information		Matrix Codes													
Company Name: Aries Clean Energy	Project Name: ASR - PFAS Study	SGS - ORLANDO Quote #: AC 11-2019-611	SKIFF #																
Address: 4037 Rural Plains Circle	Street																		
City: Franklin	State: TX	City	State																
Project Contact: Matt Newman	Email: matt.newman@ariesenergy.com	Project #																	
Phone #: 662-486-0240	Fax #: 662-471-9423	Fax #																	
Sampler(s) Name(s) (Printed): Mike Swanson / Matt Newman	Client Purchase Order #: PPO-01519																		
Sampler 1: Mike Swanson	Sampler 2: Matt Newman																		
SGS Orlando Sample #	Field ID / Point of Collection	DATE	TIME	SAMPLED BY:	MATRIX	TOTAL # OF BOTTLES	OTHER	NONE	PC	NEG	POS	P2504	NACHT-7AC	D WATER	MECH	LAB USE ONLY			
1	ASR Feed Stock	10/30/19	EDT	MS	SOL	1													
2	Cyclon Bed Oxidizing	10/16/19	EDT	MS	SOL	1													
3	Reducing Bed	10/16/19	11:20	MS	SOL	1													
4	Filter Material	10/16/19	17:25	MS	SOL	1													
Turnaround Time (Business days)		Data Deliverable Information		Comments / Remarks															
10 Day (Business) Approved By: / Date:		☐ COMMERCIAL "A" (RESULTS ONLY)		Looking to identify PFAS difference from feedstock to other materials															
7 Day		☐ COMMERCIAL "B" (RESULTS PLUS QC)																	
5 Day		☐ REDT1 (EPA LEVEL 3)																	
3 Day RUSH		☐ FULLT1 (EPA LEVEL 4)																	
2 Day RUSH		☐ EDD'S																	
1 Day RUSH																			
Other																			
Rush T/A Data Available VIA Email or Lablink		Sample Custody must be documented below each time samples change possession, including courier delivery.																	
Relinquished by Sampler/Affiliation: Mike Swanson / EERC	Date Time: 11-1-19	Received By/Affiliation: 2 Matt Newman / Aries	Relinquished By/Affiliation: 3 Matt Newman / Aries	Date Time: 11-1-19	Received By/Affiliation: 4 EERC														
Relinquished by/Affiliation	Date Time:	Received By/Affiliation	Relinquished By/Affiliation	Date Time:	Received By/Affiliation														
5		6	7		8														
Lab Use Only: Cooler Temperature (s) Celsius (corrected):		13																	

ORLD-SMT-0001-03-FORM-COC (1) Rev 031318

FA69813: Chain of Custody

Page 1 of 2



SGS Sample Receipt Summary

Job Number: FA69813

Client: ARIES CLEAN ENERGY

Project: ASR-PFAS STUDY

Date / Time Received: 11/12/2019 9:45:00 AM

Delivery Method: UPS

Airbill #'s:

Therm ID: IR 1;

Therm CF: -0.8;

of Coolers: 1

Cooler Temps (Raw Measured) °C: Cooler 1: (2.1);

Cooler Temps (Corrected) °C: Cooler 1: (1.3);

Cooler Information

Y or N

- | | | |
|-----------------------------|-------------------------------------|--------------------------|
| 1. Custody Seals Present | ☒ | ☐ |
| 2. Custody Seals Intact | ☒ | ☐ |
| 3. Temp criteria achieved | ☒ | ☐ |
| 4. Cooler temp verification | <u>IR Gun</u> | |
| 5. Cooler media | <u>Ice (Bag)</u> | |

Trip Blank Information

Y or N N/A

- | | | | |
|--------------------------------|--------------------------|--------------------------|-------------------------------------|
| 1. Trip Blank present / cooler | ☐ | ☐ | ☒ |
| 2. Trip Blank listed on COC | ☐ | ☐ | ☒ |
| | <u>W or S</u> | <u>N/A</u> | |
| 3. Type Of TB Received | ☐ | ☐ | ☒ |

Sample Information

Y or N N/A

- | | | | |
|---|-------------------------------------|-------------------------------------|-------------------------------------|
| 1. Sample labels present on bottles | ☒ | ☐ | |
| 2. Samples preserved properly | ☒ | ☐ | |
| 3. Sufficient volume/containers recvd for analysis: | ☒ | ☐ | |
| 4. Condition of sample | <u>Intact</u> | | |
| 5. Sample recvd within HT | ☒ | ☐ | |
| 6. Dates/Times/IDs on COC match Sample Label | ☒ | ☐ | |
| 7. VOCs have headspace | ☐ | ☐ | ☒ |
| 8. Bottles received for unspecified tests | ☐ | ☒ | |
| 9. Compositing instructions clear | ☐ | ☐ | ☒ |
| 10. Voa Soil Kits/Jars received past 48hrs? | ☐ | ☐ | ☒ |
| 11. % Solids Jar received? | ☐ | ☐ | ☒ |
| 12. Residual Chlorine Present? | ☐ | ☐ | ☒ |

Misc. Information

Number of Encores: 25-Gram _____ 5-Gram _____

Test Strip Lot #: pH 0-3 _____ 230315 _____

Residual Chlorine Test Strip Lot #: _____

Number of 5035 Field Kits: _____

pH 10-12 _____ 219813A _____

Number of Lab Filtered Metals: _____

Other: (Specify) _____

Comments

SM001
Rev. Date 05/24/17

Technician: PETERH

Date: 11/12/2019 9:45:00 A

Reviewer: _____

Date: _____

FA69813: Chain of Custody

Page 2 of 2

MS Semi-volatiles

5

QC Data Summaries

Includes the following where applicable:

- Method Blank Summaries
- Blank Spike Summaries
- Matrix Spike and Duplicate Summaries

Method Blank Summary

Page 1 of 2

Job Number: FA69813
Account: ARIESTNF Aries Clean Energy
Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
OP77685-MB	2Q39319.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M BY ID

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	Compound	Result	RL	MDL	Units	Q
375-22-4	Perfluorobutanoic acid	ND	1.0	0.25	ug/kg	
2706-90-3	Perfluoropentanoic acid	ND	1.0	0.20	ug/kg	
307-24-4	Perfluorohexanoic acid	ND	1.0	0.20	ug/kg	
375-85-9	Perfluoroheptanoic acid	ND	1.0	0.25	ug/kg	
335-67-1	Perfluorooctanoic acid	ND	1.0	0.25	ug/kg	
375-95-1	Perfluorononanoic acid	ND	1.0	0.25	ug/kg	
335-76-2	Perfluorodecanoic acid	ND	1.0	0.25	ug/kg	
2058-94-8	Perfluoroundecanoic acid	ND	1.0	0.25	ug/kg	
307-55-1	Perfluorododecanoic acid	ND	1.0	0.25	ug/kg	
72629-94-8	Perfluorotridecanoic acid	ND	1.0	0.25	ug/kg	
376-06-7	Perfluorotetradecanoic acid	ND	1.0	0.25	ug/kg	
375-73-5	Perfluorobutanesulfonic acid	ND	1.0	0.25	ug/kg	
2706-91-4	Perfluoropentanesulfonic acid	ND	1.0	0.25	ug/kg	
355-46-4	Perfluorohexanesulfonic acid	ND	1.0	0.25	ug/kg	
375-92-8	Perfluoroheptanesulfonic acid	ND	1.0	0.25	ug/kg	
1763-23-1	Perfluorooctanesulfonic acid	ND	1.0	0.25	ug/kg	
68259-12-1	Perfluorononanesulfonic acid	ND	1.0	0.25	ug/kg	
335-77-3	Perfluorodecanesulfonic acid	ND	1.0	0.25	ug/kg	
754-91-6	PFOSA	ND	1.0	0.25	ug/kg	
2355-31-9	MeFOSAA	ND	2.5	0.50	ug/kg	
2991-50-6	EtFOSAA	ND	2.5	0.50	ug/kg	
757124-72-44:2	Fluorotelomer sulfonate	ND	1.0	0.25	ug/kg	
27619-97-2	6:2 Fluorotelomer sulfonate	ND	1.0	0.25	ug/kg	
39108-34-4	8:2 Fluorotelomer sulfonate	ND	1.0	0.25	ug/kg	
13252-13-6	HFPO-DA (GenX)	ND	5.0	1.3	ug/kg	
919005-14-4	ADONA	ND	1.0	0.25	ug/kg	
756426-58-19	Cl-PF3ONS (F-53B Major)	ND	1.0	0.25	ug/kg	
763051-92-91	Cl-PF3OUdS (F-53B Minor)	ND	1.0	0.25	ug/kg	

CAS No.	ID Standard Recoveries	Limits
	13C4-PFBA	84% 50-150%
	13C5-PFPeA	84% 50-150%
	13C5-PFHxA	80% 50-150%
	13C4-PFHpA	81% 50-150%

Method Blank Summary

Job Number: FA69813
Account: ARIESTNF Aries Clean Energy
Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
OP77685-MB	2Q39319.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605

The QC reported here applies to the following samples: Method: EPA 537M BY ID

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	ID Standard Recoveries	Limits
13C8-PFOA	84%	50-150%
13C9-PFNA	83%	50-150%
13C6-PFDA	83%	50-150%
13C7-PFUnDA	84%	50-150%
13C2-PFDoDA	82%	50-150%
13C2-PFTeDA	86%	50-150%
13C3-PFBS	85%	50-150%
13C3-PFHxS	83%	50-150%
13C8-PFOS	86%	50-150%
13C8-FOSA	88%	50-150%
d3-MeFOSAA	87%	50-150%
13C2-4:2FTS	77%	50-150%
13C2-6:2FTS	82%	50-150%
13C2-8:2FTS	88%	50-150%
13C3-HFPO-DA	77%	50-150%

Instrument Blank

Page 1 of 2

Job Number: FA69813

Account: ARIESTNF Aries Clean Energy

Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
S2Q605-IBLK	2Q39271.D	1	11/13/19	NAF	n/a	n/a	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M QSM5.1 B-15

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	Compound	Result	RL	MDL	Units	Q
375-22-4	Perfluorobutanoic acid	ND	0.016	0.0040	ug/kg	
2706-90-3	Perfluoropentanoic acid	ND	0.016	0.0032	ug/kg	
307-24-4	Perfluorohexanoic acid	ND	0.016	0.0032	ug/kg	
375-85-9	Perfluoroheptanoic acid	ND	0.016	0.0040	ug/kg	
335-67-1	Perfluorooctanoic acid	ND	0.016	0.0040	ug/kg	
375-95-1	Perfluorononanoic acid	ND	0.016	0.0040	ug/kg	
335-76-2	Perfluorodecanoic acid	ND	0.016	0.0040	ug/kg	
2058-94-8	Perfluoroundecanoic acid	ND	0.016	0.0040	ug/kg	
307-55-1	Perfluorododecanoic acid	ND	0.016	0.0040	ug/kg	
72629-94-8	Perfluorotridecanoic acid	ND	0.016	0.0040	ug/kg	
376-06-7	Perfluorotetradecanoic acid	ND	0.016	0.0040	ug/kg	
375-73-5	Perfluorobutanesulfonic acid	ND	0.016	0.0040	ug/kg	
2706-91-4	Perfluoropentanesulfonic acid	ND	0.016	0.0040	ug/kg	
355-46-4	Perfluorohexanesulfonic acid	ND	0.016	0.0040	ug/kg	
375-92-8	Perfluoroheptanesulfonic acid	ND	0.016	0.0040	ug/kg	
1763-23-1	Perfluorooctanesulfonic acid	ND	0.016	0.0040	ug/kg	
68259-12-1	Perfluorononanesulfonic acid	ND	0.016	0.0040	ug/kg	
335-77-3	Perfluorodecanesulfonic acid	ND	0.016	0.0040	ug/kg	
754-91-6	PFOSA	ND	0.016	0.0040	ug/kg	
2355-31-9	MeFOSAA	ND	0.040	0.0080	ug/kg	
2991-50-6	EtFOSAA	ND	0.040	0.0080	ug/kg	
757124-72-44:2	Fluorotelomer sulfonate	ND	0.016	0.0040	ug/kg	
27619-97-2	6:2 Fluorotelomer sulfonate	ND	0.016	0.0040	ug/kg	
39108-34-4	8:2 Fluorotelomer sulfonate	ND	0.016	0.0040	ug/kg	
13252-13-6	HFPO-DA (GenX)	ND	0.080	0.020	ug/kg	
919005-14-4	ADONA	ND	0.016	0.0040	ug/kg	
756426-58-19	Cl-PF3ONS (F-53B Major)	ND	0.016	0.0040	ug/kg	
763051-92-91	Cl-PF3OUdS (F-53B Minor)	ND	0.016	0.0040	ug/kg	

CAS No.	ID Standard Recoveries	Limits
	13C4-PFHpA	88% 50-150%
	13C8-PFOA	92% 50-150%
	13C9-PFNA	90% 50-150%
	13C3-PFBS	93% 50-150%

Instrument Blank

Job Number: FA69813
Account: ARIESTNF Aries Clean Energy
Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
S2Q605-IBLK	2Q39271.D	1	11/13/19	NAF	n/a	n/a	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M QSM5.1 B-15

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	ID Standard Recoveries	Limits
13C3-PFHxS	91%	50-150%
13C8-PFOS	91%	50-150%

Blank Spike Summary

Page 1 of 2

Job Number: FA69813
Account: ARIESTNF Aries Clean Energy
Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
OP77685-BS	2Q39318.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M BY ID

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	Compound	Spike ug/kg	BSP ug/kg	BSP %	Limits
375-22-4	Perfluorobutanoic acid	10	10.1	101	60-134
2706-90-3	Perfluoropentanoic acid	10	9.40	94	62-134
307-24-4	Perfluorohexanoic acid	10	9.77	98	63-130
375-85-9	Perfluoroheptanoic acid	10	10.3	103	63-122
335-67-1	Perfluorooctanoic acid	10	10.2	102	71-128
375-95-1	Perfluorononanoic acid	10	9.95	100	66-124
335-76-2	Perfluorodecanoic acid	10	9.13	91	68-127
2058-94-8	Perfluoroundecanoic acid	10	9.28	93	61-137
307-55-1	Perfluorododecanoic acid	10	10.0	100	71-126
72629-94-8	Perfluorotridecanoic acid	10	9.50	95	60-137
376-06-7	Perfluorotetradecanoic acid	10	9.81	98	61-131
375-73-5	Perfluorobutanesulfonic acid	10	9.84	98	70-135
2706-91-4	Perfluoropentanesulfonic acid	10	9.29	93	70-130
355-46-4	Perfluorohexanesulfonic acid	10	10.3	103	72-129
375-92-8	Perfluoroheptanesulfonic acid	10	10.1	101	62-129
1763-23-1	Perfluorooctanesulfonic acid	10	9.98	100	69-125
68259-12-1	Perfluorononanesulfonic acid	10	9.31	93	70-130
335-77-3	Perfluorodecanesulfonic acid	10	8.98	90	63-141
754-91-6	PFOSA	10	9.58	96	65-140
2355-31-9	MeFOSAA	10	9.84	98	71-124
2991-50-6	EtFOSAA	10	11.0	110	63-129
757124-72-44:2	Fluorotelomer sulfonate	10	9.43	94	70-130
27619-97-2	6:2 Fluorotelomer sulfonate	10	9.38	94	76-131
39108-34-4	8:2 Fluorotelomer sulfonate	10	8.93	89	60-138
13252-13-6	HFPO-DA (GenX)	50	44.1	88	60-140
919005-14-4	ADONA	10	8.38	84	60-140
756426-58-19	Cl-PF3ONS (F-53B Major)	10	8.07	81	60-140
763051-92-91	Cl-PF3OUdS (F-53B Minor)	10	7.87	79	60-140

CAS No.	ID Standard Recoveries	BSP	Limits
	13C4-PFBA	90%	50-150%
	13C5-PFPeA	90%	50-150%
	13C5-PFHxA	85%	50-150%
	13C4-PFHpA	85%	50-150%

* = Outside of Control Limits.

Blank Spike Summary

Page 2 of 2

Job Number: FA69813
Account: ARIESTNF Aries Clean Energy
Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
OP77685-BS	2Q39318.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M BY ID

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	ID Standard Recoveries	BSP	Limits
	13C8-PFOA	87%	50-150%
	13C9-PFNA	88%	50-150%
	13C6-PFDA	87%	50-150%
	13C7-PFUnDA	90%	50-150%
	13C2-PFDoDA	88%	50-150%
	13C2-PFTeDA	95%	50-150%
	13C3-PFBS	91%	50-150%
	13C3-PFHxS	87%	50-150%
	13C8-PFOS	91%	50-150%
	13C8-FOSA	91%	50-150%
	d3-MeFOSAA	92%	50-150%
	13C2-4:2FTS	86%	50-150%
	13C2-6:2FTS	90%	50-150%
	13C2-8:2FTS	99%	50-150%
	13C3-HFPO-DA	88%	50-150%

* = Outside of Control Limits.

Matrix Spike/Matrix Spike Duplicate Summary

Page 1 of 2

Job Number: FA69813
Account: ARIESTNF Aries Clean Energy
Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
OP77685-MS	2Q39321.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605
OP77685-MSD	2Q39322.D	1	11/14/19	NAF	11/12/19	OP77685	S2Q605
FA69762-1	2Q39320.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M BY ID

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	Compound	FA69762-1 ug/kg	Spike Q ug/kg	MS ug/kg	MS %	Spike ug/kg	MSD ug/kg	MSD %	RPD	Limits Rec/RPD
375-22-4	Perfluorobutanoic acid	ND	12.2	12.0	98	12.1	11.9	98	1	60-134/30
2706-90-3	Perfluoropentanoic acid	ND	12.2	11.3	92	12.1	11.1	92	2	62-134/30
307-24-4	Perfluorohexanoic acid	ND	12.2	11.7	96	12.1	11.6	96	1	63-130/30
375-85-9	Perfluoroheptanoic acid	ND	12.2	12.3	101	12.1	12.1	100	2	63-122/30
335-67-1	Perfluorooctanoic acid	ND	12.2	12.1	99	12.1	12.0	99	1	71-128/30
375-95-1	Perfluorononanoic acid	ND	12.2	11.9	97	12.1	11.9	98	0	66-124/30
335-76-2	Perfluorodecanoic acid	ND	12.2	10.8	88	12.1	10.8	89	0	68-127/30
2058-94-8	Perfluoroundecanoic acid	ND	12.2	11.1	91	12.1	11.0	91	1	61-137/30
307-55-1	Perfluorododecanoic acid	ND	12.2	11.9	97	12.1	11.7	97	2	71-126/30
72629-94-8	Perfluorotridecanoic acid	ND	12.2	12.0	98	12.1	11.8	97	2	60-137/30
376-06-7	Perfluorotetradecanoic acid	ND	12.2	11.7	96	12.1	11.6	96	1	61-131/30
375-73-5	Perfluorobutanesulfonic acid	ND	12.2	11.8	97	12.1	11.5	95	3	70-135/30
2706-91-4	Perfluoropentanesulfonic acid	ND	12.2	11.0	90	12.1	10.9	90	1	70-130/30
355-46-4	Perfluorohexanesulfonic acid	ND	12.2	12.3	101	12.1	12.0	99	2	72-129/30
375-92-8	Perfluoroheptanesulfonic acid	ND	12.2	10.6	87	12.1	11.6	96	9	62-129/30
1763-23-1	Perfluorooctanesulfonic acid	0.391	J 12.2	14.1	112	12.1	12.0	96	16	69-125/30
68259-12-1	Perfluorononanesulfonic acid	ND	12.2	11.9	97	12.1	10.5	87	13	70-130/30
335-77-3	Perfluorodecanesulfonic acid	ND	12.2	12.5	102	12.1	10.8	89	15	63-141/30
754-91-6	PFOSA	ND	12.2	11.4	93	12.1	11.2	93	2	65-140/30
2355-31-9	MeFOSAA	ND	12.2	11.3	92	12.1	11.3	93	0	71-124/30
2991-50-6	EtFOSAA	ND	12.2	13.6	111	12.1	13.4	111	1	63-129/30
757124-72-44:2	Fluorotelomer sulfonate	ND	12.2	11.2	92	12.1	11.0	91	2	70-130/30
27619-97-2	6:2 Fluorotelomer sulfonate	ND	12.2	11.2	92	12.1	11.1	92	1	76-131/30
39108-34-4	8:2 Fluorotelomer sulfonate	ND	12.2	10.6	87	12.1	10.6	88	0	60-138/30

CAS No.	ID Standard Recoveries	MS	MSD	FA69762-1	Limits
	13C4-PFBA	76%	89%	88%	50-150%
	13C5-PFPeA	75%	88%	88%	50-150%
	13C5-PFHxA	72%	84%	83%	50-150%
	13C4-PFHpA	70%	82%	82%	50-150%
	13C8-PFOA	64%	85%	86%	50-150%
	13C9-PFNA	59%	84%	85%	50-150%
	13C6-PFDA	73%	85%	86%	50-150%
	13C7-PFUnDA	73%	86%	87%	50-150%

* = Outside of Control Limits.

Matrix Spike/Matrix Spike Duplicate Summary

Page 2 of 2

Job Number: FA69813

Account: ARIESTNF Aries Clean Energy

Project: PFAS Analysis

Sample	File ID	DF	Analyzed	By	Prep Date	Prep Batch	Analytical Batch
OP77685-MS	2Q39321.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605
OP77685-MSD	2Q39322.D	1	11/14/19	NAF	11/12/19	OP77685	S2Q605
FA69762-1	2Q39320.D	1	11/13/19	NAF	11/12/19	OP77685	S2Q605

The QC reported here applies to the following samples:

Method: EPA 537M BY ID

FA69813-1, FA69813-2, FA69813-3, FA69813-4

CAS No.	ID Standard Recoveries	MS	MSD	FA69762-1	Limits
13C2-PFDoDA		76%	88%	87%	50-150%
13C2-PFTeDA		78%	92%	92%	50-150%
13C3-PFBS		77%	90%	89%	50-150%
13C3-PFHxS		75%	88%	87%	50-150%
13C8-PFOS		67%	89%	88%	50-150%
13C8-FOSA		66%	77%	81%	50-150%
d3-MeFOSAA		78%	93%	92%	50-150%
13C2-4:2FTS		73%	85%	80%	50-150%
13C2-6:2FTS		63%	85%	83%	50-150%
13C2-8:2FTS		89%	104%	100%	50-150%
13C3-HFPO-DA		0%*	0%*		50-150%

* = Outside of Control Limits.